

Total No. of Questions : 8]

SEAT No. :

P796

[Total No. of Pages : 2

**[5870] - 1110**  
**T.E. (Chemical)**  
**CHEMICAL TECHNOLOGY - II**  
**(2019 Pattern) (Semester - I) (309342)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates :*

- 1) Answer Q.No.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Assume suitable data if necessary.*
- 4) Use of logarithmic tables, slide rule, Mollier charts, electronic pocket calculator and steam tables is permitted.*
- 5) Figures to the right indicate full marks.*

**Q1) a)** Explain the process of manufacturing of Portland cement with major engineering problems. **[9]**

b) Describe various steps involved in pyrolysis of coal. **[8]**

OR

**Q2) a)** Explain carbonization of coal and the effect of temperature on the products obtained. **[9]**

b) Explain construction and working of Blast furnace. **[8]**

**Q3) a)** Describe different types of solvents used in paint manufacture? **[9]**

b) Explain production of water gas and its applications. **[9]**

OR

**Q4) a)** Write a short note on the following with suitable diagram (any 2) **[12]**

i) Producer Gas

ii) Natural Gas

iii) Water Gas

b) Write the classification of dyes with their properties. **[6]**

**P.T.O.**

**Q5) a)** Discuss different types of agrochemicals and explain manufacturing process of pesticides. **[9]**

b) Discuss properties and applications of Varnishes. **[8]**

OR

**Q6) a)** Explain manufacturing of penicillin with major engineering problems. **[9]**

b) Describe in detail about plant growth regulators and yield stimulators. **[8]**

**Q7) a)** Explain production of methanol in brief with a neat process flow diagram. **[9]**

b) Describe production of styrene with its engineering problems. **[9]**

OR

**Q8)** Draw a neat flow diagram and explain in brief (any 3) **[18]**

a) Production of vinyl chloride

b) Production of Phenol

c) Production of acetone

d) Production of Ethylene



Total No. of Questions : 8]

SEAT No. :

PA-1409

[Total No. of Pages : 2

[5926]-25

T.E. (Chemical)

CHEMICAL ENGINEERING MATHEMATICS

(2019 Pattern) (309343) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6 and Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data, if necessary.

Q1) a) Use least-squares regression to fit a straight line to : [8]

|   |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|
| x | 1   | 2   | 3   | 4   | 5   | 6   | 7   |
| y | 0.5 | 2.5 | 2.0 | 4.0 | 3.5 | 6.0 | 5.5 |

b) Use a Lagrange interpolating polynomial of the second order to evaluate  $f(2)$  on the basis of the data given : [9]

|      |   |       |       |
|------|---|-------|-------|
| x    | 1 | 4     | 6     |
| F(x) | 0 | 1.386 | 1.792 |

OR

Q2) a) Use least-squares regression to fit a parabola to : [10]

|   |    |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|----|
| x | 5  | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 |
| y | 17 | 24 | 31 | 33 | 37 | 37 | 40 | 40 | 42 | 41 |

b) Find the integration of  $(3x + 4)$  in the limits 2 to 5 by Trapezoidal rule using six steps. [7]

Q3) a) Solve the following initial value problem over the interval from  $t = 0$  to  $t = 2$

where  $y(0) = 1$  using Euler's method with  $h = 0.25$   $\frac{dy}{dx} = yt^2 - 2y$ . [10]

b) Given that  $\frac{dy}{dx} = x + y^2$  and  $y = 0$  when  $x = 0$ , determine the value of  $y$  when  $x = 0.3$ , correct to four places of decimals using Picards method. [8]

OR

P.T.O.

**Q4) a)** Solve the following problem using second order RK method from  $t = 0$  to 1 taking  $h = 0.5$   $\frac{dy}{dt} = -y + t^2$   $y(0) = 1$ . [10]

**b)** Obtain the approximate solution  $y(t)$  of IVP using Taylor series method. Obtain approximate solution at  $t = 0.1$  correct to 4 places of decimal  $\frac{dy}{dt} = 1 + ty$   $y(0) = 1$ . [8]

**Q5)** Use the explicit method to solve the temperature distribution of a long thin rod with a length of 10 cm and the following values :  $k' = 0.49 \text{ cal/(s.cm. } ^\circ\text{C)}$ ,  $\Delta x = 2 \text{ cm}$  and  $\Delta t = 0.1 \text{ s}$ . At  $t = 0$ , the temperature of the rod is zero and the boundary conditions are fixed for all times at  $T(0) = 100^\circ\text{C}$  and  $T(10) = 50^\circ\text{C}$ . The rod is of aluminium with  $C = 0.2174 \text{ cal/g. } ^\circ\text{C}$  and  $\rho = 2.7 \text{ g/cm}^3$ . Therefore,  $k = 0.49/(2.7 * 0.2174) = 0.835 \text{ cm}^2/\text{s}$  and  $\lambda = 0.835(0.1)/(2)^2 = 0.020875$ . Do two iterations. [18]

$$T_i^{l+1} = T_i^l + \lambda (T_{i+1}^l - 2T_i^l + T_{i-1}^l) \text{ where } \lambda = k \frac{\Delta t}{\Delta x^2}$$

OR

**Q6) a)** Explain implicit method and its advantages and disadvantages in details. [12]  
**b)** Explain classification of partial differential equations with examples. [6]

**Q7) a)** Use Newton's method to approximate the maximum of  $f(x) = 2 \sin x - (x^2 / 10)$  with initial guess  $x_0 = 1$ . Do three iterations. [10]  
**b)** What are the six steps of optimization? Enlist and explain. [7]

OR

**Q8)** Using Simplex method, Maximize  $Z = 5x_1 + 3x_2$  [17]

Subject to:  $x_1 + x_2 \leq 2$

$$5x_1 + 2x_2 \leq 10$$

$$3x_1 + 8x_2 \leq 12$$

$$x_1, x_2 \geq 0$$

